

Package Insert

1. NAME OF THE MEDICINAL PRODUCT

Seacross Palonosetron 250 micrograms/5ml Solution for injection

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each ml of solution contains 50 micrograms palonosetron (as hydrochloride).

Each vial of 5 ml of solution contains 250 micrograms palonosetron (as hydrochloride).

0.28mg/5 ml Palonosetron (as hydrochloride) is equivalent to 0.25mg/5 ml Palonosetron

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Solution for injection.

Clear, colourless solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Palonosetron is indicated for:

Chemotherapy-Induced Nausea and Vomiting

Adults and Paediatric Patients 1 month of Age and Older

- the prevention of acute nausea and vomiting associated with highly emetogenic cancer chemotherapy.
- the prevention of nausea and vomiting associated with moderately emetogenic cancer chemotherapy.

Postoperative Nausea and Vomiting

- Prevention of postoperative nausea and vomiting (PONV) for up to 24 hours following surgery. Efficacy beyond 24 hours has not been demonstrated.

As with other antiemetics, routine prophylaxis is not recommended in patients in whom there is little expectation that nausea and/or vomiting will occur postoperatively. In patients where nausea and vomiting must be avoided during the postoperative period, Palonosetron is recommended even where the incidence of postoperative nausea and/or vomiting is low.

4.2 Posology and method of administration

This medicinal product should be administered by a healthcare professional under appropriate medical supervision.

Posology

Adults

Chemotherapy-Induced Nausea and Vomiting

250 micrograms palonosetron administered as a single intravenous bolus approximately 30 minutes before the start of chemotherapy. Palonosetron should be injected over 30 seconds.

The efficacy of Palonosetron in the prevention of nausea and vomiting induced by highly emetogenic chemotherapy may be enhanced by the addition of a corticosteroid administered prior to chemotherapy.

Postoperative Nausea and Vomiting

75 micrograms palonosetron administered as a single intravenous bolus over 10 seconds immediately before the induction of anaesthesia

Paediatric population

Chemotherapy-Induced Nausea and Vomiting

Children and Adolescents (aged 1 month to 17 years):

20 micrograms/kg (the maximum total dose should not exceed 1500 micrograms) palonosetron administered as a single 15 minute intravenous infusion beginning approximately 30 minutes before the start of chemotherapy.

The safety and efficacy of Palonosetron Hydrochloride in children aged less than 1 month have not been established. No data are available.

Postoperative Nausea and Vomiting

Safety and effectiveness in patients below the age of 18 years have not been established.

Elderly people

No dose adjustment is necessary for the elderly.

Hepatic impairment

No dose adjustment is necessary for patients with impaired hepatic function.

Renal impairment

No dose adjustment is necessary for patients with impaired renal function.

No data are available for patients with end stage renal disease undergoing haemodialysis.

Method of administration

For intravenous use.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

4.4 Special warnings and precautions for use

Hypersensitivity

Hypersensitivity reactions may occur in patients who have exhibited hypersensitivity to other 5-HT₃ receptor antagonists.

Serotonin Syndrome

Symptoms associated with serotonin syndrome may include the following combination of signs and symptoms: mental status change (e.g. agitation, hallucinations, delirium, and coma), autonomic instability (e.g. tachycardia, labile blood pressure, dizziness, diaphoresis, flushing, hyperthermia), neuromuscular symptoms (e.g. tremor, rigidity, myoclonus, hyperreflexia, incoordination), seizures, with or without gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea). Patients should be monitored for the emergence of serotonin syndrome, especially with concomitant use of Palonosetron hydrochloride solution for injection and other serotonergic drugs (e.g selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), monoamine oxidase inhibitors, mirtazapine, fentanyl, lithium, tramadol and intravenous methylene blue). If symptoms of serotonin syndrome occur, discontinue Palonosetron hydrochloride solution for injection and initiate supportive treatment. Patients should be informed of the increased risk of serotonin syndrome, especially if Palonosetron hydrochloride solution for injection is used concomitantly with other serotonergic drugs (see Section Interactions with Other Medicaments).

4.5 Interaction with other medicinal products and other forms of interaction

Palonosetron is eliminated from the body through both renal excretion and metabolic pathways with the latter mediated via multiple CYP enzymes. Palonosetron is not an inhibitor of CYP1A2, CYP2A6, CYP2B6, CYP2C9, CYP2D6, CYP2E1 and CYP3A4/5 nor does it induce the activity of CYP1A2, CYP2D6, or CYP3A4/5. Therefore, the potential for clinically significant drug interactions with palonosetron appears to be low.

Chemotherapeutic agents

Palonosetron did not inhibit the antitumour activity of the five chemotherapeutic agents tested (cisplatin, cyclophosphamide, cytarabine, doxorubicin and mitomycin C).

Corticosteroids

Palonosetron has been administered safely with corticosteroids.

Serotonergic Drugs (e.g. SSRIs and SNRIs)

Serotonin syndrome (including altered mental status, autonomic instability, and neuromuscular symptoms) has been described following the concomitant use of 5-HT₃ receptor antagonists and other serotonergic drugs, including selective serotonin reuptake inhibitors (SSRIs) and serotonin and noradrenaline reuptake inhibitors (SNRIs) (see Section Warnings and Precautions).

Antiemetic / Antinauseants

No significant interaction between palonosetron and oral aprepitant and oral metoclopramide.

Other medicinal products

Palonosetron has been administered safely with analgesics, antispasmodics and anticholinergic medicinal products.

4.6 Fertility, pregnancy and lactation

Pregnancy

Teratogenic Effects: Category B

No adequate and well-controlled studies in pregnant women. Therefore, palonosetron should be used during pregnancy only if clearly needed

Labor and Delivery

Palonosetron has not been administered to patients undergoing labor and delivery, so its effect on the mother or child are unknown.

Breast-feeding

It is not known whether palonosetron is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants and the potential for tumorigenicity shown for palonosetron, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. Since palonosetron may induce dizziness, somnolence or fatigue, patients should be cautioned when driving or operating machines.

4.8 Undesirable effects

Chemotherapy-Induced Nausea and Vomiting

Adults

The most common adverse reactions were headache and constipation. The following adverse reactions were reported for palonosetron.

System organ class	1%	<1%
Cardiovascular	Non-sustained Tachycardia, bradycardia, hypotension	Hypertension, myocardial ischaemia, extrasystoles, sinus tachycardia, sinus arrhythmia, supraventricular extrasystoles and QT prolongation
Dermatological		Allergic dermatitis, rash
Hearing and Vision		Motion sickness, tinnitus, eye irritation and amblyopia.
Gastrointestinal System	Diarrhoea	Dyspepsia, abdominal pain, dry mouth, hiccups, flatulence
General	Weakness	Fatigue, fever, hot flush, flu-like syndrome
Liver		Transient, asymptomatic increases in AST and / or ALT and bilirubin. These changes occurred predominantly in patients receiving highly emetogenic chemotherapy.
Metabolism	Hyperkalaemia	Electrolyte fluctuations, hyperglycemia, metabolic acidosis, glycosuria, appetite decrease, anorexia
Musculoskeletal		Arthralgia
Nervous system	Dizziness	Somnolence, insomnia, paraesthesia, hypersomnia
Psychiatric	Anxiety	Euphoric mood
Urinary system		Urinary retention
Vascular disorders		Vein discolouration, vein distended

Paediatric population

The following adverse reactions were reported for palonosetron.

System organ class	1%	<1%
Nervous system		Headache, Dizziness, dyskinesia
General		Infusion site pain
Dermatological		Allergic dermatitis, skin disorder

Postoperative Nausea and Vomiting

Adults

The following adverse reactions were reported for palonosetron.

System organ class	1%	<1%
Cardiovascular	Electrocardiogram QTc prolongation, sinus bradycardia, tachycardia	Blood pressure decreased, hypotension, hypertension, arrhythmia, ventricular extrasystoles, generalized edema; ECG T wave amplitude decreased, platelet count decreased.
Dermatological	Pruritus	
Gastrointestinal System	Flatulence	Dry mouth, upper abdominal pain, salivary hypersecretion, dyspepsia, diarrhea, intestinal hypomotility, anorexia
General		Chills
Liver	Increase in AST and / or ALT	Hepatic enzyme increased
Metabolism		Hypokalemia, anorexia

Nervous system		Dizziness
Respiratory		Hypoventilation, laryngospasm
Urinary system	Urinary retention	

Post marketing Experience

Very rare cases (<1/10000): Hypersensitivity reactions and injection site reactions (burning, induration, discomfort and pain).

4.9 Overdose

There is no known antidote to Palonosetron hydrochloride solution for injection. Overdose should be managed with supportive care.

Doses of up to 6 mg have been used in adult. Due to the large volume of distribution, dialysis is unlikely to be an effective treatment for palonosetron overdose. The major signs of toxicity were convulsions, gasping, pallor, cyanosis and collapse.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antiemetics and antinauseants, serotonin (5HT₃) antagonists. ATC code: A04AA05

Mechanism of action

Palonosetron is a selective high-affinity receptor antagonist of the 5HT₃ receptor.

Palonosetron is a 5-HT₃ receptor antagonist with a strong binding affinity for this receptor and little or no affinity for other receptors.

Cancer chemotherapy may be associated with a high incidence of nausea and vomiting, particularly when certain agents, such as cisplatin, are used. 5-HT₃ receptors are located on the nerve terminals of the vagus in the periphery and centrally in the chemoreceptor trigger zone of the area postrema. It is thought that chemotherapeutic agents produce nausea and vomiting by releasing serotonin from the enterochromaffin cells of the small intestine and that the released serotonin then activates 5-HT₃ receptors located on vagal afferents to initiate the vomiting reflex.

Postoperative nausea and vomiting is influenced by multiple patient, surgical and anesthesia related factors and is triggered by release of 5-HT in a cascade of neuronal events involving both the central nervous system and the gastrointestinal tract. The 5-HT₃ receptor has been demonstrated to selectively participate in the emetic response

The effect of palonosetron on blood pressure, heart rate, and ECG parameters including QTc were comparable to ondansetron and dolasetron in CINV. In PONV, the effect of palonosetron on the QTc interval was no different from placebo. Palonosetron possesses the ability to block ion channels involved in ventricular de- and re-polarisation and to prolong action potential duration.

No significant effect on any ECG interval including QTc duration (cardiac repolarization) at doses up to 2.25 mg.

5.2 Pharmacokinetic properties

Absorption

Following intravenous administration, an initial decline in plasma concentrations is followed by slow elimination from the body. Mean maximum plasma concentration (C_{max}) and area under the

concentration-time curve ($AUC_{0-\infty}$) are generally dose-proportional over the dose range of 0.3–90 µg/kg.

After intravenous dosing of palonosetron in patients undergoing surgery (abdominal surgery or vaginal hysterectomy), the pharmacokinetics characteristics of palonosetron were similar to those observed in cancer patients

Distribution

Palonosetron has a volume of distribution of approximately 8.3 +/- 2.5 L/kg. Approximately 62 % of palonosetron is bound to plasma proteins.

Metabolism

Palonosetron is eliminated by multiple routes with approximately 50% metabolised to form two primary metabolites: N-oxide-palonosetron and 6-S-hydroxy-palonosetron. These metabolites each have less than 1 % of the 5HT₃ receptor antagonist activity of palonosetron. It was shown that CYP2D6 and to a lesser extent, CYP3A4 and CYP1A2 are involved in the metabolism of palonosetron. However, clinical pharmacokinetic parameters are not significantly different between poor and extensive metabolisers of CYP2D6 substrates.

Elimination

After a single intravenous dose of 10 micrograms/kg [14C]-palonosetron, approximately 80 % of the dose was recovered within 144 hours in the urine with palonosetron representing approximately 40 % of the administered dose. Mean terminal elimination half-life is approximately 40 hours.

Special populations

Paediatric Patients

No apparent differences in volume of distribution when expressed in L/kg.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Mannitol
Disodium edetate
Sodium citrate
Citric acid monohydrate
Sodium hydroxide (for pH adjustment)
Hydrochloric acid (for pH adjustment)
Water for injections

6.2 Incompatibilities

This medicinal product must not be mixed with other medicinal products.

6.3 Shelf life

4 years.

Upon opening of the vial, use immediately and discard any unused solution. Protect from light. For single use only.

6.4 Storage Condition

Do not store above 30°C. Protect from light. For single use only.

6.5 Pack Size

1 x 5ml vials

6.6 Instruction for use

Single use only, any unused solution should be discarded.

Any unused product or waste material should be disposed of in accordance with local requirements

7. PRODUCT REGISTRATION HOLDER

First Pharmaceutical Sdn Bhd
No. 20 Jalan SS 19/5,
47500 Subang Jaya, Selangor, Malaysia

8. MANUFACTURER

Sichuan Huiyu Pharmaceutical Company Limited
Building 3, No. 333, Hanyang Road,
Shizhong District, Neijiang, CN-641000, China

9. MARKETING AUTHORISATION NUMBER

MAL23076015ACZ

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